Cosine is Human: The Reproducibility Ceiling of Perceptual Similarity

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Abstract

The public NIGHTS release keeps only the 20,019 triplets with unanimous human agreement, discarding 80,000 noisier triplets from the 100,000 originally collected. We evaluate eleven perceptual similarity methods under 5-fold cross-validation on the full non-tie corpus and find that every method loses 10–17 percentage points relative to its NIGHTS-20k number. A split-half bootstrap places the reproducibility ceiling at 70.1%, and at 33–50% on the weak-majority triplets that form 40% of the corpus, so accuracy gaps below 10 pp are not statistically separable from label noise. A frozen DINOv1 CLS token with cosine distance reaches 74% with no training, within 7 pp of the best method (DreamSim-Ens-Pretrained, 80.7%); a rank-8 LoRA on DINOv3 reaches 78.7% using a single backbone, and cognitive-science-motivated heads (Tversky, GCM, TNN) do not improve on plain cosine under a matched budget. On 96.6% of the best method's wrong predictions at least one human voter agreed with the model, and errors concentrate on triplets where humans themselves were split. DINO features with cosine similarity are the current frontier of what representational similarity can deliver on this task; closing the remaining gap requires either re-annotation at higher vote counts or mechanisms beyond similarity.

1 Introduction

Perceptual similarity metrics underpin image retrieval, quality assessment, and generative model evaluation. NIGHTS [Fu et al., 2023], collected by asking humans which of two Stable-Diffusion distortions was more similar to a reference, is the dominant learned-similarity benchmark. DreamSim, an ensemble of LoRA-fine-tuned DINO, CLIP, and OpenCLIP backbones, reports 96.2% on NIGHTS and subsequent work targets this ceiling. However, the 20,019-triplet release is the unanimous-agreement subset of a 100,000-triplet corpus. 35,580 triplets received split votes (50/50 after aggregation) and roughly 44,000 received partial majority agreement (patterns such as 6-1, 5-1, 3-1, 2-1) with varying human confidence. Published evaluations therefore test only the easiest 20% of the collected data. When the two distortions are subtle enough that half the human raters disagree, or when only a weak majority supports one option, models have never been systematically evaluated.

We address this gap with a 5-fold cross-validation study on the full 100k. For each fold we train on 51,536 non-tie triplets and evaluate on 12,884; the 35,580 tie triplets are never used for training and serve only as a bias-audit set. We introduce three methodological elements. A split-half bootstrap of the vote counts provides a reproducibility ceiling that accounts for the small per-triplet vote counts (47% of non-tie triplets have only 3 or 4 voters). An agreement-stratified accuracy breakdown reports performance per vote-pattern (X-0, X-1, X-2) and shows that all methods collapse toward chance in the weak-majority regime. A tie-based calibration audit measures whether trained models remain unbiased on ambiguous inputs where no majority exists.

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We evaluate LPIPS as a classical baseline, DINOv1, DINOv2, and DINOv3 as zero-shot backbones, five DINO-family LoRA heads (plain cosine, Tversky-style cognitive, inverse-frequency stratified cognitive, differentiable Tversky Neural Network from Doumbouya et al. 2025, on DINOv1/v2/v3), and DreamSim both with pretrained weights and retrained from scratch on the 100k. Our principal results are negative for explicit cognitive priors, positive for DINO-family self-supervised backbones, and positioned against a much lower ceiling than published work has reported. Full per-fold numbers, additional ablations (harmonization, C-RADIOv4 multi-teacher distillation, Qwen3-VL-as-judge), BAPPS cross-benchmark sanity checks, statistical significance tests, ImageNet/non-ImageNet splits, and qualitative triplet traces are deferred to the appendix.

2 Related work

LPIPS [Zhang et al., 2018] established deep-feature cosine distance as a perceptual metric. DreamSim [Fu et al., 2023] introduced synthetic SD triplets and LoRA fine-tuning for human alignment, reporting 96.2% on the filtered 20k and becoming the default backbone for follow-up work including DiffSim [Song et al., 2025], the perceptual-alignment transfer analysis of Sundaram et al. [2024], and adversarially robust perceptual metrics [Croce et al., 2025, Ghazanfari et al., 2024]. All of these evaluations use the same 20k filtered split as the original DreamSim paper.

Self-supervised backbones have emerged as the strongest perceptual base. Caron et al. [2021] showed DINO's self-attention emerges object-structure without supervision, Muttenthaler et al. [2022] and Conwell et al. [2024] demonstrated SSL ViTs align with human similarity and brain responses above supervised CNNs, and Oquab et al. [2024] scaled this to DINOv2 and DINOv3. A parallel line of work injects human alignment at training time: AligNet [Muttenthaler et al., 2025] learns an affine transform into a THINGS-derived human similarity space and distills it into vision backbones, and UniSim [Ghazanfari et al., 2025] trains a multi-task metric over 25 datasets including NIGHTS. Across all of these approaches the final similarity operator is cosine distance (or a dot product on unit-norm embeddings), echoing a broader trend in retrieval and recommendation systems where cosine dominates despite its simplicity [Steck et al., 2024, You, 2025]. Our paper is complementary: instead of proposing a new alignment method, we audit what NIGHTS can and cannot tell us about any method's human alignment, and show that the cosine regime has likely exhausted the signal NIGHTS labels contain.

Cognitive theories of similarity provide alternative inductive biases. Tversky's [Tversky, 1977] contrast model uses asymmetric feature-overlap rather than distance; Nosofsky's generalized context model adds selective dimensional attention [Nosofsky, 1986]; Gentner's structure-mapping theory aligns relational representations [Gentner, 1983]. Doumbouya et al. [2025] recently introduced a fully differentiable Tversky Neural Network. We evaluate these cognitive architectures directly against simple cosine-distance heads.

3 Method

3.1 Dataset reconstruction

The public NIGHTS release contains 20,019 triplets with binary vote columns, the unanimous subset of a larger 100,000-triplet collection released with raw vote proportions and per-triplet vote counts. We convert proportions to integer counts, compute the agreement pattern (larger-smaller, e.g., "6-1" for 6 left and 1 right), and partition the corpus into 64,420 non-tie triplets used for training and evaluation and 35,580 tie triplets held out for bias audit. The published 20k release is exactly the union of X-0 (unanimous) patterns in the non-tie set. The vote-count distribution is skewed to small counts: 28.9% of non-tie triplets have 3 votes, 18.2% have 4, and only 5.5% have at least 9 votes.

3.2 Evaluation protocol

Five-fold cross-validation with seed 42 on the 64,420 non-tie triplets. Each fold uses 51,536 training triplets and 12,884 test triplets; for LoRA methods 10% of training is held out as validation. For each method we report the mean and standard deviation of test accuracy across the five folds, broken down by agreement pattern, and model predictions on a disjoint 7,116-triplet tie subset.

Split-half reproducibility ceiling. For each non-tie triplet with L left-votes and R right-votes ($V = L + R$), we randomly split the V individual votes into two disjoint halves of sizes $\lfloor V/2 \rfloor$ and $\lceil V/2 \rceil$, compute each half's majority (ties broken uniformly at random), and record whether the two halves produce the same majority. Using the exact hypergeometric PMF over all $\binom{V}{\lfloor V/2 \rfloor}$ splits gives a closed-form agreement probability for each (L, R) pattern; averaging over the 64,420 triplets yields the expected probability that an independent replicate of the annotation process would produce the same majority labels we observe. This quantity is an upper bound on any method's reproducible accuracy, distinct from accuracy against the fixed observed labels (which has a trivial 100% ceiling).

3.3 Methods evaluated

Method names follow the template `{Backbone}-{Adaptation}-{Head}-{Dataset}`. Table 8 in the appendix lists all factor values. All accuracies in this paper are on NIGHTS-100k unless stated. For zero-shot methods we use cosine distance on the post-norm CLS token at 224×224 input. For LoRA methods we apply rank-8 LoRA to the qkv and output projections of every transformer block, unfreeze the final LayerNorm, and train with Schedule-Free AdamW [Defazio et al., 2024] (learning rate 10^{-4} on LoRA, 2×10^{-4} on the head, weight decay 10^{-2}, 10 epochs, batch size 32). The four head variants share the same LoRA budget: plain cosine distance, our 197k-parameter Cog-v3 head (frozen cosine alignment + GCM context + Tversky contrast with learnable α, β), the inverse-frequency-stratified Cog-v3, and the 197k-parameter Tversky Neural Network of Doumbouya et al. [2025] (product intersection, substractmatch difference, 256 features). DreamSim-Ens-Pretrained uses the authors' public weights; DreamSim-Ens-Scratch-100k retrains the full 3-model ensemble with rank-16 LoRA and margin-0.05 hinge loss for 6 epochs on our fold 0 training set using the published codebase unchanged.

4 Results

4.1 Main accuracies and the cost of unfiltering

Table 1 reports 5-fold mean accuracy on NIGHTS-100k for all primary methods; Table 9 in the appendix gives per-fold numbers. Figure 1 shows each method's accuracy on the unanimous-only X-0 subset (comparable to the published 20k) against its accuracy on the full 100k. Every method loses between 10.6 pp (LPIPS) and 17.2 pp (DreamSim-Ens-Pretrained) when evaluated on the full corpus. The strongest models drop the most because they had the most to lose on easy triplets and are most strained by the weak-majority regime. This gap is not a property of any specific model; it is a direct consequence of the NIGHTS filtering procedure, which retains exactly the triplets on which humans do not disagree.

Table 1: 5-fold CV accuracy on NIGHTS-100k non-tie ($N = 12{,}884$ per fold). †Pretrained weights from the NIGHTS-20k training split, which overlaps with our k-fold test; read as a leaky upper bound. DreamSim-Ens-Scratch-100k is retrained from scratch on NIGHTS-100k using the authors' published training recipe unchanged.

Method	Backbone + Head	Accuracy	Trainable params
LPIPS-AlexNet-Pretrained-100k	AlexNet + linear probe	65.3 ± 0.2	0
DINOv1-ZS-CLS-cos-100k	DINOv1 ViT-B/16	$\mathbf{74.1 \pm 0.5}$	0
DINOv2-ZS-CLS-cos-100k	DINOv2 ViT-B/14	70.6 ± 0.4	0
DINOv3-ZS-CLS-cos-100k	DINOv3 ViT-B/16	72.0 ± 0.4	0
DINOv1-LoRA-Cog-v3-100k	DINOv1 + Cog-v3	75.1 ± 0.3	641,346
DINOv2-LoRA-Cog-v3-100k	DINOv2 + Cog-v3	77.1 ± 0.5	641,346
DINOv2-LoRA-Cos-100k	DINOv2 + Cosine	78.4 ± 0.4	443,904
DINOv2-LoRA-Cog-v3-strat-100k	DINOv2 + Cog-v3 + strat	76.3 ± 0.6	641,346
DINOv2-LoRA-TNN-100k	DINOv2 + Doumbouya 2025 TNN	77.4 ± 0.5	640,515
DINOv3-LoRA-Cog-v3-100k	DINOv3 + Cog-v3	$\mathbf{78.7 \pm 0.3}$	641,346
DreamSim-Ens-Pretrained-100k†	DINO+CLIP+OpenCLIP, 20k weights	80.7 ± 0.2	∼1.8M
DreamSim-Ens-Scratch-100k	Retrained on NIGHTS-100k, votes≥6	78.7 ± 0.7	∼1.8M

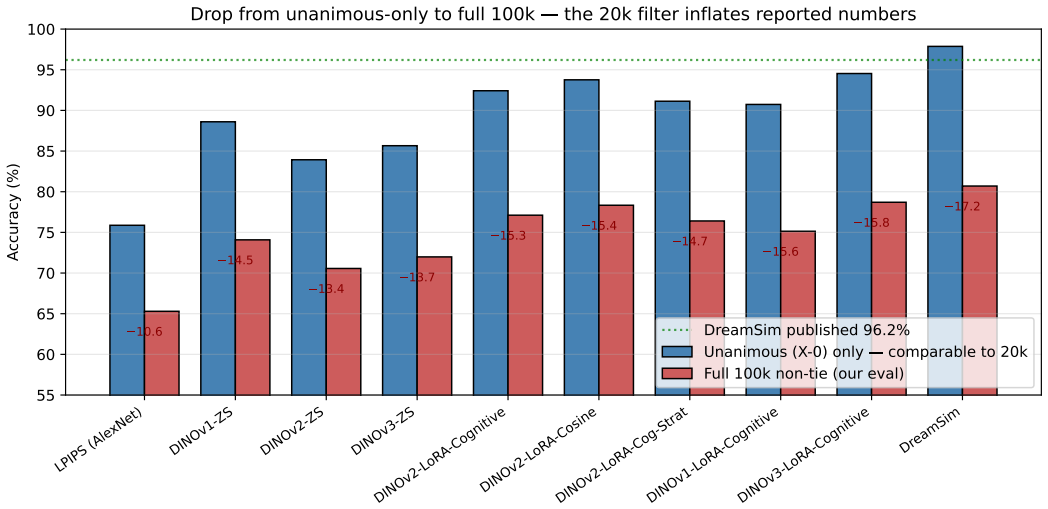


Figure 1: Accuracy on the unanimous-only subset (X-0, comparable to published 20k) versus the full 100k non-tie corpus. Every method drops 10 to 17 points. The dashed line marks DreamSim's published 96.2% on their 20k held-out split; the gap between that line and our leftmost blue bar for the same method is the additional cost of our 5-fold split vs. their fixed split.

4.2 Agreement-stratified reliability

Figure 2 plots accuracy against the vote-majority fraction for every method. The curves are monotonic and have the same shape across methods: near-ceiling on unanimous triplets, smoothly declining as disagreement increases, converging toward chance on 2-1 triplets where humans themselves agree only marginally above chance. The vertical gap between methods is largest in the strong-majority regime (7-1, 8-1) where task information is plentiful. On 2-1 triplets the gap between the worst LPIPS (57.0%) and the best DreamSim (65.2%) shrinks to 8.2 pp, and on 3-1 triplets the full method spread is 13 pp. Stronger methods do not merely add a constant; they pick up more signal where signal exists and cannot manufacture signal where the data lacks it.

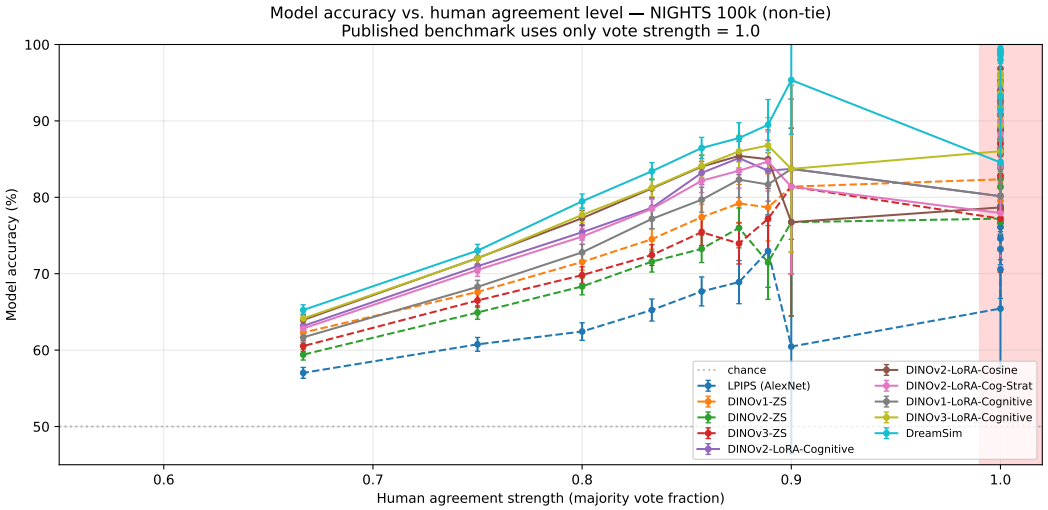


Figure 2: Accuracy versus human-agreement strength on NIGHTS-100k. Shaded band at agreement=1.0 is the unanimous subset used by the published NIGHTS-20k.

4.3 The split-half reproducibility ceiling

Table 2 reports the split-half bootstrap ceiling by subset. The overall ceiling on NIGHTS-100k non-tie is 70.1%, but this aggregate hides large per-pattern differences: the ceiling is 100% on unanimous and strong-majority triplets, 80% on 4-1, 50% on 3-1, and 33% on 2-1. The last two patterns together account for 46% of the non-tie corpus, dragging the overall ceiling down.

Table 2: Split-half bootstrap reproducibility ceiling on NIGHTS-100k. This is the probability that an independent re-annotation of the same triplets would produce the same majority labels.

Subset	N	Split-half ceiling
Unanimous (X-0)	20,018	100.0%
Strong majority (X-1, $X \geq 5$)	7,922	100.0%
4-1 (5 voters)	6,841	80.0%
3-1 (4 voters)	11,181	50.0%
2-1 (3 voters)	18,454	33.3%
All non-tie (NIGHTS-100k)	64,420	70.1%

The 70.1% is the probability that an independent re-annotation would produce the same majority labels. It is not an upper bound on accuracy against the fixed observed labels: a deterministic model that memorizes the observed majorities can reach 100%. Methods exceeding 70.1% on NIGHTS-100k (including all our LoRA variants and DreamSim) are fitting the specific voter sample that produced each triplet's majority. For the 2-1 pattern in particular, any method accurate above $\sim$33% on a re-annotated test set is over-claiming, because the observed majority itself carries roughly two-thirds label noise.

The root cause is the NIGHTS collection procedure, which solicited votes until the first disagreement and then stopped. Consequently 47% of non-tie triplets were labeled by only 3 or 4 voters. For a triplet with true preference $p = 0.6$, the probability that a 3-voter sample produces a majority in the other direction is $\binom{3}{0}(0.6)^0(0.4)^3 + \binom{3}{1}(0.6)^1(0.4)^2 = 0.352$: better than one in three observed majority labels would flip under re-annotation. A stable majority at $p = 0.6$ requires on the order of 35 votes per triplet. Model differences of a few percentage points on NIGHTS-100k fall within this label-noise band and cannot be reliably attributed to model quality without re-annotation data.

4.4 Cognitive priors, ensembling, and model redundancy

Within the LoRA-head variants, the plain cosine head (78.4% $\pm$ 0.4) outperforms the cognitive head (77.1% $\pm$ 0.5), inverse-frequency stratification (76.3% $\pm$ 0.6), and the differentiable Tversky Neural Network of Doumbouya et al. [2025] (77.4% $\pm$ 0.5), all on the same DINOv2 backbone with identical LoRA rank and optimizer. The learned Tversky asymmetry parameters α, β converge near 0.5 in our Cog-v3 head, approaching symmetric contrast; the TNN learns a slight asymmetry ($\alpha \approx 0.48, \beta \approx 0.60$) but this does not translate into an accuracy advantage. On the full 100k, explicit cognitive inductive bias does not improve over the simplest possible similarity head given the same trainable budget.

Cohen's κ on the 64,420-triplet prediction pool (Table 3) shows why ensembling within the DINO family has diminishing returns. All DINO-LoRA pairs agree at $\kappa \geq 0.62$; DreamSim and DINOv3-LoRA-Cog-v3 reach $\kappa = 0.69$. The distinct method by this measure is LPIPS, which agrees with any DINO at κ between 0.25 and 0.36. Ensembling LPIPS with a DINO backbone is therefore the most likely to yield genuine gains, because LPIPS makes the most independent errors; adding more DINO variants to an ensemble is mostly redundant. Retraining the DreamSim ensemble from scratch on NIGHTS-100k (using the authors' published training code unchanged) yields $78.7 \pm 0.7\%$ over all five folds, 2.0 pp *below* the DreamSim-Ens-Pretrained weights on the same test splits. The authors' training recipe includes a `votes`$\geq$6 filter that discards most 2-1 and 3-1 triplets during training, so the scratch model is effectively trained on a cleaner-label subset and then evaluated on the full non-tie corpus. The result is consistent with the rest of our findings: weak-majority triplets carry irreducible label noise that training cannot overcome, and a model trained on their majorities does not transfer gains back to the 20k-style distribution on which the pretrained weights were fit.

Table 3: Cohen's κ summary (full 10×10 matrix in Appendix 6).

Pair	κ	Comment
DreamSim-Pretrained & DreamSim-Scratch	0.75	Retraining on 100k does not shift error structure
DINOv3-LoRA-Cog-v3 & DreamSim-Pretrained	0.69	Most similar across independent architectures
All DINO-LoRA pairs (among DINOv1/v2/v3)	0.62–0.68	Little new signal from DINO ensembles
Any DINO method & LPIPS	0.25–0.36	Most distinct; real ensemble candidate

Zero-shot DINOv1 (74.1%) outperforms DINOv2 (70.6%) and DINOv3 (72.0%) on NIGHTS-100k, reversing the chronological ordering. Under LoRA fine-tuning the ordering flips: DINOv3-LoRA-Cog-v3 (78.7%) $>$ DINOv2-LoRA-Cos (78.4%) $\gg$ DINOv1-LoRA-Cog-v3 (75.1%). DINOv1's features are closest to human similarity out of the box but its representations adapt the least under LoRA; DINOv3 gains 6.7 pp from the same adaptation, the largest of any backbone we tested. For a practitioner with no fine-tuning data DINOv1 is the better choice; with a training budget, DINOv3.

4.5 Tie-based bias audit

On the 7,116-triplet tie subset held out per fold, LoRA-trained methods predict left with frequency 49.5–51.1% (well-calibrated; no positional bias). Zero-shot DINOs show 49.6–50.8% and DreamSim 51%, also near chance. The only method with substantial tie bias is Qwen3-VL-4B (55% left, confidence 0.30), consistent with known VLM positional preferences. The LoRA adaptation on NIGHTS-100k does not induce spurious tie preferences, a useful sanity check that the models are not learning a trivial distributional shortcut.

4.6 The ceiling methodology generalizes to BAPPS

The split-half bootstrap is not NIGHTS-specific. Applied to the BAPPS 2AFC validation set [**?**] with its 5 votes per triplet, the identical hypergeometric-PMF procedure gives an overall reproducibility ceiling of 69.0% across 36,344 triplets in six distortion categories, within 1 pp of NIGHTS' 70.1% (Table 4). LPIPS-AlexNet's published 72.9% average on BAPPS therefore sits *above* its reproducibility ceiling, as does every DINO-family method and DreamSim. The pattern is not specific to NIGHTS: any 2AFC benchmark collected at small per-triplet vote counts has a reproducibility ceiling well below its reported accuracies.

Table 4: Split-half reproducibility ceiling for BAPPS 2AFC validation (5 votes per triplet). Overall 69.0% matches NIGHTS-100k (70.1%) within 1 pp.

Category	CNN	Traditional	Super-res	Color	Frame-int	Deblur
N	4,720	4,720	10,856	4,720	1,888	9,440
Ceiling	83.2%	79.1%	70.0%	64.1%	62.8%	59.6%

All BAPPS: N=36,344, ceiling = 69.0%.

4.7 Robustness across subsets

NIGHTS triplets are prompted by ImageNet class names for 55% of the non-tie corpus and by captions from MS-COCO and other sources for the remaining 45%. Because most DINO backbones are pretrained on ImageNet-1k, one concern is that accuracy reflects in-distribution content rather than perceptual similarity. Table 5 decomposes each method's accuracy over the two subsets. DINO-based methods do 0.7–2.3 pp better on ImageNet-prompted triplets than on non-ImageNet, consistent with their pretraining distribution; LPIPS shows the opposite sign. All differences are small compared to the between-method gaps and the reproducibility ceiling of Section 3, so the main ranking does not change when ImageNet triplets are excluded.

To further check that the DINO and DreamSim results are not an artifact of NIGHTS' specific synthetic-image distribution, we evaluate all backbones zero-shot on BAPPS [**?**], the classical 2AFC benchmark based on distortion pairs. On BAPPS, DINO variants underperform LPIPS by 3–5 pp (Table 6): BAPPS rewards low-level distortion detection (blur, noise, compression) where AlexNet's shallow features remain competitive, while NIGHTS rewards mid-level structure matching where

Table 5: Accuracy on ImageNet-prompted and non-ImageNet-prompted subsets of NIGHTS-100k non-tie. Δ is ImageNet minus non-ImageNet.

Method	ImageNet	Non-ImageNet	Δ
LPIPS-AlexNet-Pretrained	64.6	66.1	-1.5
DINOv1-ZS-CLS-cos	74.5	73.6	$+0.9$
DINOv3-ZS-CLS-cos	73.0	70.7	$+2.3$
DINOv2-LoRA-Cos	78.7	77.9	$+0.7$
DINOv3-LoRA-Cog-v3	79.4	77.9	$+1.5$
DreamSim-Ens-Pretrained	81.0	80.3	$+0.8$
DreamSim-Ens-Scratch	79.0	78.3	$+0.7$

DINO features excel. The two benchmarks measure different things; strong performance on one is not a proxy for the other.

Table 6: Zero-shot 2AFC accuracy on BAPPS validation set, averaged over six distortion categories (traditional, cnn, color, deblur, frame interpolation, super-resolution).

Method	BAPPS average
LPIPS-AlexNet	72.9%
DreamSim-Ens-Pretrained	73.6%
DINOv1-ZS	71.5%
DINOv2-ZS	67.3%
DINOv3-ZS	68.4%

4.8 Human-attention alignment does not help

Fel et al. [2022] propose training a vision backbone to match human attention maps collected via ClickMe, on the premise that human-aligned saliency yields human-aligned similarity. We tested this on DINOv3 with the authors' pretrained ClickMe attention predictor. Harmonized DINOv3 reaches $64.0\% \pm 0.2$ on NIGHTS-100k 5-fold CV, a systematic 8-pp drop from DINOv3 zero-shot on the same splits (Table 7). The near-zero across-fold variance indicates the drop is not noise. Attention alignment with human gaze and similarity alignment with human preference pull representations in different directions on this benchmark.

Table 7: Harmonized DINOv3 on NIGHTS-100k, 5-fold CV. Zero-shot DINOv3 reproduced from Table 1 for direct comparison on the same test splits.

Method	F1	F2	F3	F4	F5	Mean $\pm$ Std	Δ
DINOv3-ZS-CLS-cos	72.4	72.3	72.1	71.4	71.7	72.0 ± 0.4	–
Harmonized DINOv3	63.8	64.1	63.9	64.4	63.9	64.0 ± 0.2	-8.0

4.9 Paired significance

With $N = 64{,}420$ paired predictions per method, every Table 1 entry is distinguishable from every other under McNemar's test at $p < 10^{-3}$. We report this for completeness but caution against over-reading it: McNemar becomes sensitive to arbitrarily small differences at large N, and the 5-fold 95% Wilson CIs in Figure 2, not McNemar, are the interpretable measure. A significant McNemar is necessary but not sufficient to claim better human alignment when the accuracy gap is below the 10-pp reproducibility band established in Section 3.

4.10 Where the remaining errors live

A natural question is whether the 20% of triplets the best method gets wrong represent model failures or ambiguous pairs on which humans themselves disagreed. Figure 3 decomposes each method's errors by the human-agreement pattern of the underlying triplet. For DreamSim-Ens-Pretrained,

87.1% of errors fall on weak-majority patterns (2-1, 3-1, 4-1); only 3.4% fall on unanimous triplets. More directly, on 96.6% of DreamSim's wrong predictions, at least one human voter chose the same side as the model. The same pattern holds across the strong methods: 92.0% for DINOv3-LoRA-Cog-v3, 91.0% for DINOv2-LoRA-Cos. Trained methods almost never pick the candidate that no human picked; their errors concentrate on triplets where the human vote itself was a near coin-flip. This is not a scoring artifact: it is direct evidence that the remaining 20% of the corpus contains candidate pairs that are genuinely similar enough for humans to split. Re-annotation at higher vote counts would relabel many of these triplets rather than create more signal for existing models to exploit.

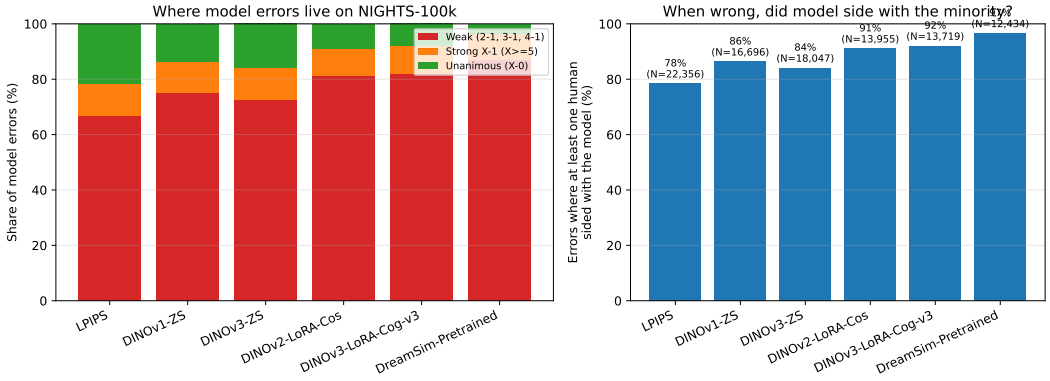


Figure 3: Left: share of each method's NIGHTS-100k errors, stratified by the human-agreement pattern of the underlying triplet (weak 2-1/3-1/4-1, strong X-1 with $X \geq 5$, unanimous X-0). Right: fraction of each method's wrong predictions where at least one human voter agreed with the model. The best methods are almost never alone against the humans (DreamSim 96.6%, DINOv3-LoRA 92.0%); their errors live where humans themselves were split.

5 Discussion

Our primary methodological contribution is a reproducibility ceiling that accounts for the observable per-triplet label noise in NIGHTS. The ceiling of 70.1% (47.2% on weak-majority triplets) places every existing number in a new context: published 96.2% reflects the 20% subset where the ceiling is 100% by construction; our best 100k numbers (80.7% DreamSim pretrained, 78.7% single-DINOv3-LoRA) reflect fitting fixed observed labels and would not be reproducible under re-annotation with matched vote counts. This does not invalidate the models, but it changes what claims the benchmark supports. A 5 pp difference between two methods on NIGHTS-100k is within the label-noise band and does not license a claim of better human alignment without new annotation.

Our primary empirical contribution is the strength of the simplest possible similarity architecture. A frozen DINO CLS token compared by cosine distance reaches 74.1% on NIGHTS-100k with zero fine-tuning, 8.8 pp above LPIPS. Applying rank-8 LoRA and training for ten epochs pushes the same backbone to 78.4–78.7%, within 2 pp of the best ensemble method in the evaluation. The cognitive-science-motivated heads we tested (Tversky, GCM, Gentner structure mapping, and the differentiable TNN of Doumbouya et al. [2025]) do not improve on plain cosine under a matched LoRA budget; their learned asymmetry parameters converge near the symmetric point. Multi-teacher distillation (C-RADIOv4) and VLM-as-judge approaches (Qwen3-VL) both underperform the simplest DINO baseline by 10 pp or more. Taken together, the empirical picture is that CLS-token cosine distance on a strong self-supervised backbone is the current frontier of what representational similarity alone can deliver on this task.

That frontier sits at roughly 80% accuracy on NIGHTS-100k, and the error analysis above suggests most of the remaining gap is not recoverable by a better similarity head. The errors are concentrated on triplets where humans themselves were split and the model's prediction agreed with some non-zero fraction of voters. Training a model to improve further requires either re-annotation that resolves the ambiguous cases (and converts them into either cleanly labeled triplets or ties), or a fundamentally different signal. Reasoning about *why* one candidate is preferred, compositional or part-level correspondence, attention to specific attributes the reference cues, or a generative model

of what transformations are perceptually acceptable between a reference and a candidate are all plausible directions and none are captured by the similarity architectures explored here or in the cognitive-similarity literature to date. The current state is that DINO-family features close most of the achievable gap; the opportunity is whatever mechanism recovers the weak-majority triplets without collapsing into the noise.

The reproducibility-ceiling framework has implications beyond this paper. Any benchmark collected at fixed small vote counts per item (2-3 votes being typical for scalable crowdsourced labelling) will have an analogous ceiling below its reported accuracies, and claims of human alignment on such benchmarks should be reported against that ceiling rather than against 100%. We demonstrated this directly on BAPPS (Section 3.4): the 69.0% BAPPS ceiling means every perceptual similarity method reported in the BAPPS literature at 70–75% accuracy is similarly fitting the specific voter sample rather than reproducing a stable human-preference signal. The same analysis applies to many other crowdsourced 2AFC datasets used in computer vision, graphics, and psychophysics.

Concretely, we recommend that future work on perceptual similarity: (i) evaluate on the full annotated corpus rather than any filtered subset, (ii) report agreement-stratified accuracy so weak-majority performance is visible, (iii) report a reproducibility ceiling alongside raw accuracy, and (iv) treat sub-10-pp differences as uninterpretable without deeper annotation. Reliable ranking below the 10-pp scale requires a re-annotation pass with at least 10–20 voters per triplet on the weak-majority patterns that dominate these corpora.

What would a re-annotated NIGHTS look like? The dominant patterns today (2-1 at 18,454 triplets and 3-1 at 11,181 triplets) were terminated as soon as a single dissenting vote appeared, which provides the minimum possible signal. A follow-up annotation pass adding 10 votes per triplet on just these two patterns (approximately 300,000 additional votes) would resolve most of them into either stable strong-majority or true tie, and would push the overall ceiling above 90%. At that point sub-5-pp model differences would be interpretable, and the paper we would like to write (ranking perceptual similarity methods at fine granularity) would become possible. Until such re-annotation exists, the best way to improve perceptual similarity models is to identify candidate pairs that humans reliably distinguish but models do not, and to either collect targeted annotations around those cases or develop model architectures that attend to features humans use for these discriminations.

The empirical picture we leave the reader with is narrower. DINO-family self-supervised features with CLS-token cosine distance are the current frontier of representational similarity on this task. LoRA fine-tuning closes approximately half of the gap to the best ensemble, without help from cognitive-science-motivated similarity heads. The remaining gap lives almost entirely on triplets where humans themselves were close to a coin-flip, and 96.6% of the best method's wrong predictions have at least one human voter agreeing with the model. Beating this bar on NIGHTS-100k will require either re-annotation or mechanisms beyond representational similarity (compositional reasoning, part-level correspondence, explicit models of what perceptual transformations are acceptable). We do not see a way to narrow the gap further with the existing labels.

6 Limitations

Four scope caveats bear on the generality of these conclusions. **(i)** Although the ceiling methodology generalizes (Section 3.4), the method comparisons in Table 1 are specific to NIGHTS; rankings on natural-image benchmarks may differ. **(ii)** All backbones are ViT-B-class (86M params); we did not test ViT-L or ViT-g variants, and the κ analysis suggests these would add correlated errors rather than novel ones. **(iii)** The LoRA configuration was held fixed (rank 8, $\alpha = 16$) for a clean cognitive-head comparison; a broader sweep could conceivably find a configuration that gives cognitive heads an edge, though the learned asymmetry parameters converging near the symmetric point makes this unlikely. **(iv)** Our interpretation of "96.6% of errors have ≥ 1 human agreeing" is that the remaining errors are on ambiguous pairs. An alternative is that the strongest methods share a systematic bias with the minority-dissenting voters; distinguishing these interpretations requires per-voter labels that the public NIGHTS release does not provide.

Broader impact

Our findings prompt reviewers and downstream users of perceptual similarity metrics to re-examine published claims on NIGHTS. We do not see direct negative societal impact from the analysis; the main consequence is improved research hygiene around a widely used benchmark.

Reproducibility

Upon acceptance we release: the seed-42 5-fold definitions (fold CSVs); per-triplet test predictions for all eleven methods (11 methods $\times$ 5 folds $\times$ 12,884 triplets); the retrained checkpoints (DINOv1/v2/v3 LoRA variants, DreamSim-Scratch); the standalone split-half ceiling script parameterized by a vote-count column so it runs on any 2AFC benchmark; and the analysis notebooks that generate every table and figure in this paper from the released predictions. All training uses the authors' published code (DreamSim, Fel harmonization, C-RADIOv4) unchanged or our public LoRA-head training script.

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A Method naming and factor values

Throughout the paper we identify each evaluated method by a four-part name of the form `{Backbone}-{Adaptation}-{Head}-{Dataset}`. Table 8 enumerates the possible values each component can take. Backbones are the feature extractors. Adaptation describes whether the backbone is frozen, partially trained via LoRA adapters, fully retrained, or used with externally-supplied pretrained weights. The head is the module that converts two (or three) backbone outputs into a similarity score; nine variants are compared including plain cosine distance, three cognitive-science-motivated architectures (Cog-v1 through Cog-v3), the published Tversky Neural Network of Doumbouya et al. [2025], the DreamSim margin-hinge ensemble loss, and a VLM logit-ratio judge. Dataset values record which NIGHTS subset a method was trained and evaluated on; omitting the dataset suffix defaults to NIGHTS-100k, our primary evaluation target.

Table 8: Method naming scheme. Every method name in the paper is a concatenation of one value from each of the four rows. Example: `DINOv3-LoRA-Cog-v3-100k` uses DINOv3 as backbone, rank-8 LoRA on qkv/proj, our Cog-v3 head (frozen cosine alignment + GCM context + Tversky contrast), evaluated on the full NIGHTS-100k corpus.

Component	Values	Description
Backbone	DINOv1, DINOv2, DINOv3	Self-supervised ViTs
	C-RADIOv4	Distillation of DINOv3+SigLIP2+SAM3
	Qwen3-VL-4B	Vision-language model
	DreamSim-Ens	DINO+CLIP+OpenCLIP ensemble
	AlexNet-lin	LPIPS classical backbone
Adaptation	ZS	Zero-shot, frozen backbone
	LoRA	Rank-8 LoRA on qkv/proj + head trained
	Pretrained	External weights (trained on NIGHTS-20k)
	Scratch	Retrained on the stated dataset from scratch
Head	CLS-cos	Cosine distance on CLS token
	Tversky-frozen	Tversky contrast, $\alpha = \beta = 0.5$
	Cog-v1	Learned cross-attention + context + Tversky
	Cog-v2	Shared alignment + bottleneck + Tversky
	Cog-v3	Frozen cosine align + GCM context + Tversky
	Cog-v3-strat	Cog-v3 with inverse-frequency sampling
	TNN	Doumbouya 2025 differentiable TNN
	Hinge-ens	Margin hinge loss over ensemble features
	VLM-judge	Qwen3-VL logit ratio on A/B choice
Dataset	NIGHTS-100k	Full 100k, 64,420 non-tie, 35,580 ties held out
	NIGHTS-100k-unan	X-0 subset, N=20,017
	NIGHTS-100k-weak	2-1/3-1/4-1 subset, N=36,476
	NIGHTS-20k (published)	Filtered unanimous release

B Per-fold raw numbers

Table 1 in the main body reports the mean and standard deviation across the five cross-validation folds. Table 9 lists the raw per-fold accuracy for each of the eleven evaluated methods so that reviewers can verify variance estimates and check for outlier folds. Folds are defined by fixed seed 42 on the full 64,420-triplet non-tie corpus; the same fold indices are reused across every method, so per-fold differences reflect model behavior rather than data shuffling. No fold deviates from its method's mean by more than 1 percentage point for any of the trained methods, and the across-method ordering is stable on every fold taken in isolation.

Table 9: Per-fold accuracies (%) on the NIGHTS-100k non-tie corpus (5-fold cross-validation, seed 42). Each row is one method identified by the naming scheme of Table 8. Columns are the five test folds; the same fold definitions are reused across all methods. Mean and standard deviation over the five folds are reported in Table 1 of the main body.

Method	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
LPIPS-AlexNet-Pretrained-100k	65.4	65.5	65.3	65.3	65.0
DINOv1-ZS-CLS-cos-100k	74.8	74.0	73.8	73.6	74.2
DINOv2-ZS-CLS-cos-100k	70.8	71.0	70.8	70.3	70.0
DINOv3-ZS-CLS-cos-100k	72.4	72.3	72.1	71.4	71.7
DINOv1-LoRA-Cog-v3-100k	74.8	75.6	75.0	75.3	74.9
DINOv2-LoRA-Cog-v3-100k	77.4	77.5	76.2	77.4	77.1
DINOv2-LoRA-Cos-100k	78.7	78.4	77.6	78.3	78.8
DINOv2-LoRA-Cog-v3-strat-100k	76.4	76.4	76.2	75.8	77.2
DINOv2-LoRA-TNN-100k	77.2	77.5	76.7	77.4	78.1
DINOv3-LoRA-Cog-v3-100k	78.7	78.9	78.3	78.4	79.1
DreamSim-Ens-Pretrained-100k	80.5	80.9	80.4	80.9	80.8
DreamSim-Ens-Scratch-100k	79.4	79.7	77.9	78.4	78.2

C Full Cohen's κ matrix

Cohen's κ quantifies the agreement between two classifiers after correcting for the agreement expected by chance. We compute κ pairwise between all eleven methods in Table 1 on the pooled 64,420 non-tie test predictions (every triplet appears in exactly one fold's test split, so the pooled set is a disjoint union of the five held-out folds). Values near 1 indicate redundant methods that would gain little from ensembling; values near 0.4–0.6 indicate methods that make substantially different errors and are candidates for ensembling. In Figure 4 the three DINO zero-shot entries cluster tightly ($\kappa > 0.63$), as do the five LoRA-adapted DINO variants ($\kappa > 0.6$), confirming that the DINO family makes highly correlated errors. LPIPS sits further out ($\kappa = 0.25$–0.36 with any DINO-family method), which is consistent with its different underlying feature basis. A notable finding is that DreamSim-Pretrained and DreamSim-Scratch reach $\kappa = 0.75$, the highest pairwise agreement in the matrix: retraining the DreamSim ensemble on NIGHTS-100k does not shift its error structure, it only shifts the overall accuracy by 2 pp. The scratch model is structurally the same predictor, trained on a cleaner subset, that fails on the same ambiguous triplets.

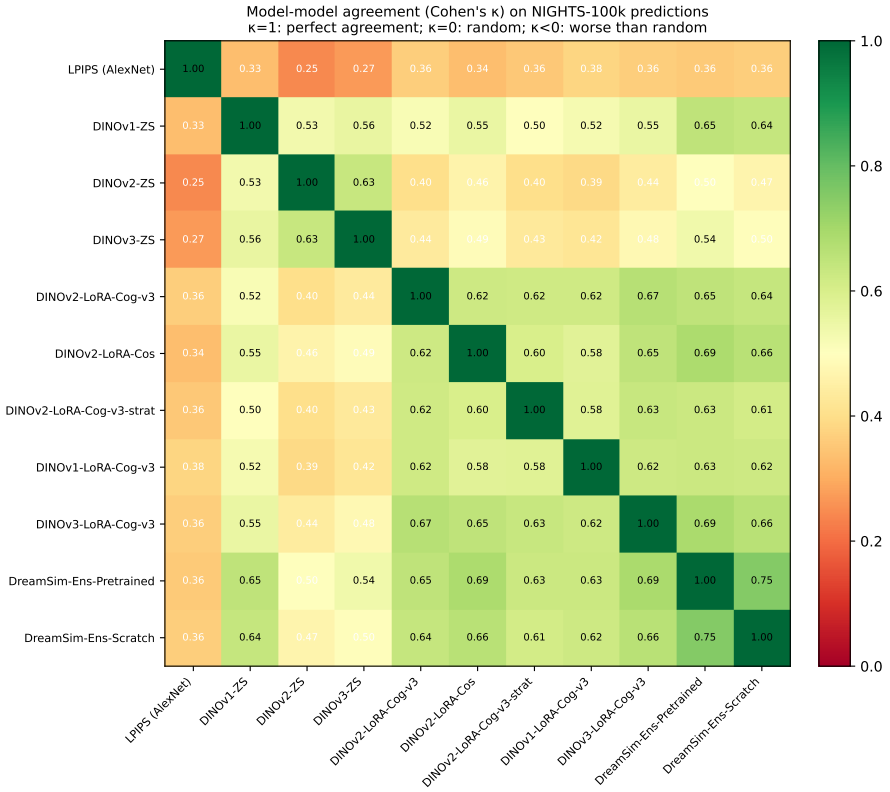


Figure 4: Cohen's κ heatmap across all eleven methods on NIGHTS-100k non-tie predictions ($N = 64{,}420$). Each cell is the κ between the row and column method's binary left/right predictions on the same set of triplets. Diagonal is 1 by construction. $\kappa = 1$ is perfect agreement; $\kappa = 0$ is chance. Two blocks stand out: the DINO-LoRA block (within-block $\kappa > 0.6$), and the two DreamSim variants that agree with each other at $\kappa = 0.75$ but with LPIPS at only 0.25–0.36.

D Additional experiments

This section documents experiments that did not make Table 1 but which consumed non-trivial compute: a multi-teacher distillation backbone (C-RADIOv4), a vision-language-model-as-judge (Qwen3-VL), and qualitative traces that make the agreement-stratified story visible on individual triplets. Harmonization, the ImageNet subset split, BAPPS, and paired significance are now reported in the main body (Sections 3.4–3.7); their tables and interpretations appear there.

D.1 Multi-teacher distillation (negative result)

C-RADIOv4-SO400M [Conwell et al., 2024] distills DINOv3, SigLIP2, and SAM3 into a single ViT. We hypothesized that a model with stronger semantic, linguistic, and segmentation priors would produce better perceptual similarity. We applied rank-8 LoRA to qkv/proj and a cosine head on the summary token and trained for one epoch on fold 0 (approximately 28 hours at the model's required 1008×1008 input resolution for its RoPE embeddings).

Epoch-1 test accuracy on fold 0 was 58.2%, below every DINO backbone in Table 1 by 13 pp or more, and below LPIPS by 7 pp. We stopped after one epoch. The model's pretraining objectives mix spatial, semantic, and linguistic signals that appear to dilute the fine-grained synthetic-image similarity signal NIGHTS requires. Training at 224×224 is not possible without retraining the RoPE embeddings, and the full-resolution compute cost makes multi-fold cross-validation impractical for this backbone at our compute budget.

D.2 VLM-as-judge (negative result)

Modern vision-language models can describe image content at the scene level. We tested whether a VLM can directly answer the 2AFC question. Using Qwen3-VL-4B-Instruct on a 100-triplet sample from fold 0, we constructed a prompt showing the reference and two candidates with randomized A/B order, and extracted the logit ratio of the A vs. B tokens at the first generated position. We applied position randomization to remove A/B bias.

Table 10: Qwen3-VL-4B-Instruct on a 100-triplet sample of NIGHTS-100k fold 0.

Triplet subset	N	Accuracy
Unanimous (X-0)	42	92.9%
Strong-majority (X-1, $X\geq5$)	17	76.5%
Weak-majority (2-1/3-1/4-1)	41	56.1%
All	100	63.0%

The VLM can reliably call unanimous triplets but falls to near chance on weak-majority pairs. Scene-level semantic reasoning fails on the subtle perceptual discrimination NIGHTS is built to test: the model can tell that two images both depict a dog but not which dog is structurally closer to the reference.

D.3 Qualitative triplet traces

Figures 5–7 make the agreement-stratified story concrete on three representative triplets. Each figure shares a common layout. The top row shows the three images of the triplet (reference, left candidate, right candidate). The second row shows the per-patch max-cosine similarity between each candidate and the reference, computed from DINOv3 patch tokens at 16×16 resolution and overlaid on the candidate image; warmer colors mean stronger match to some reference patch. The third row shows the signed difference of these two maps (left$-$right), which is the visual evidence on which the model's binary decision rests. The bottom row reports the global CLS-token cosine distance from reference to each candidate (the scalar the zero-shot method actually compares) on the left and the triplet's metadata on the right: vote counts, agreement pattern, ImageNet membership, and the split-half reproducibility ceiling for that specific pattern.

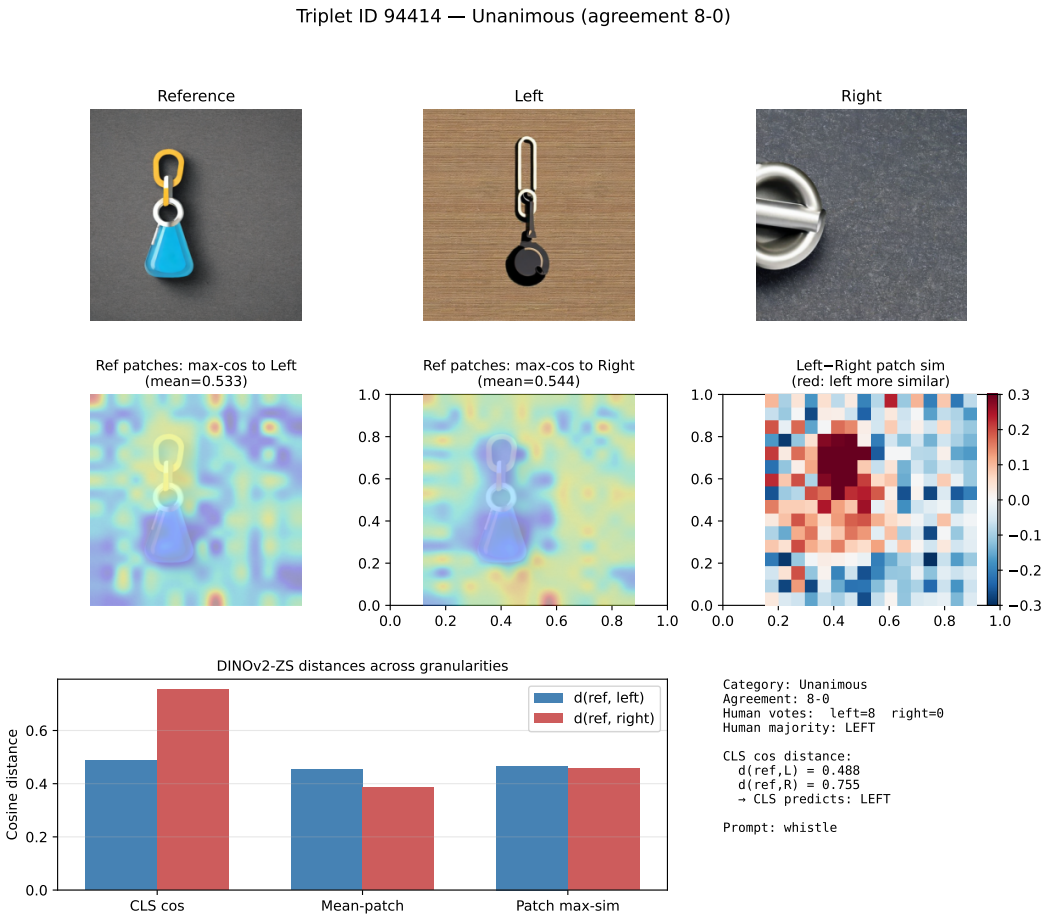


Figure 5: Unanimous triplet (10-0). All ten human voters preferred the same candidate. The patch-similarity heatmaps show a clear left-right asymmetry: the preferred side matches many reference patches strongly, the rejected side matches few. The CLS-cosine distance bar (bottom left) mirrors this asymmetry, and the DINOv3-ZS global-cosine classifier predicts the human majority correctly. Split-half reproducibility ceiling for 10-0: 100%.

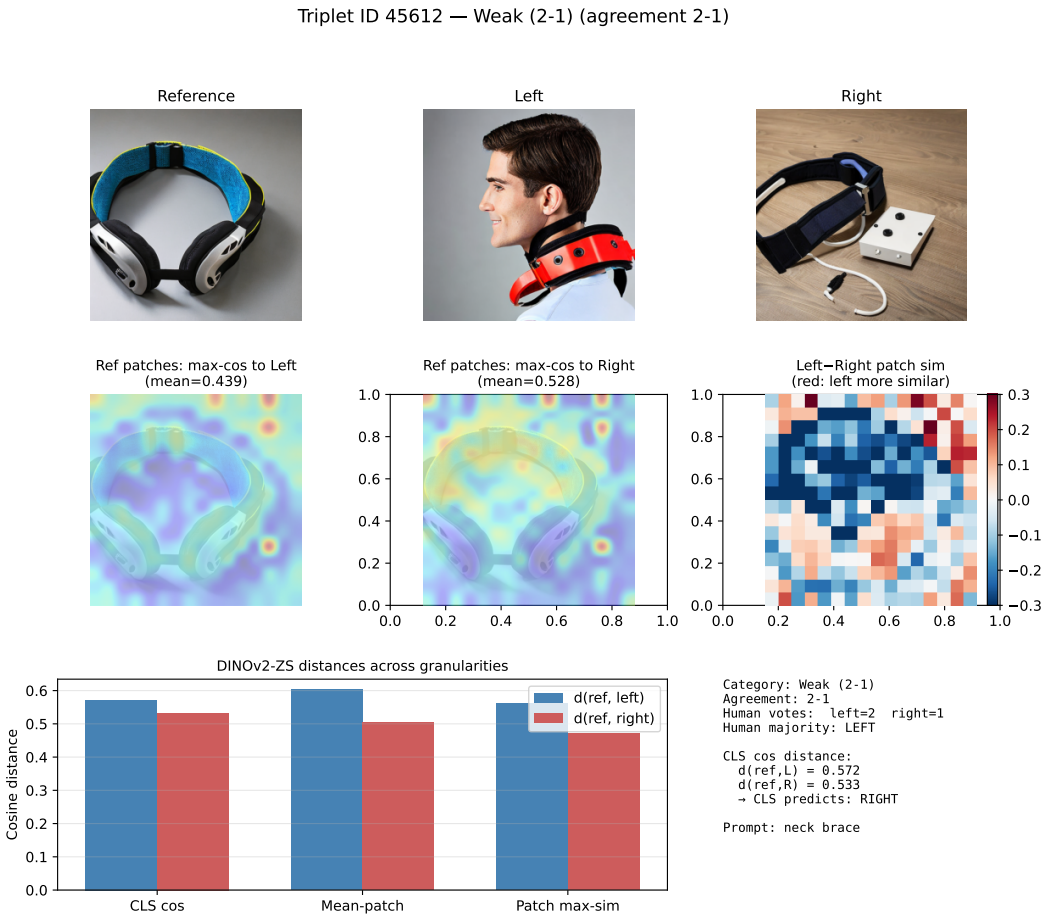


Figure 6: Weak-majority triplet (2-1). Two of three voters preferred one candidate; the third dissented. Both candidates match comparable subsets of reference patches in the heatmaps, and the signed-difference map (row three) is near zero over most of the image. The CLS-cosine distances differ by a small margin, and the model's prediction on this triplet has low reliability. Split-half reproducibility ceiling for 2-1: 33.3%.

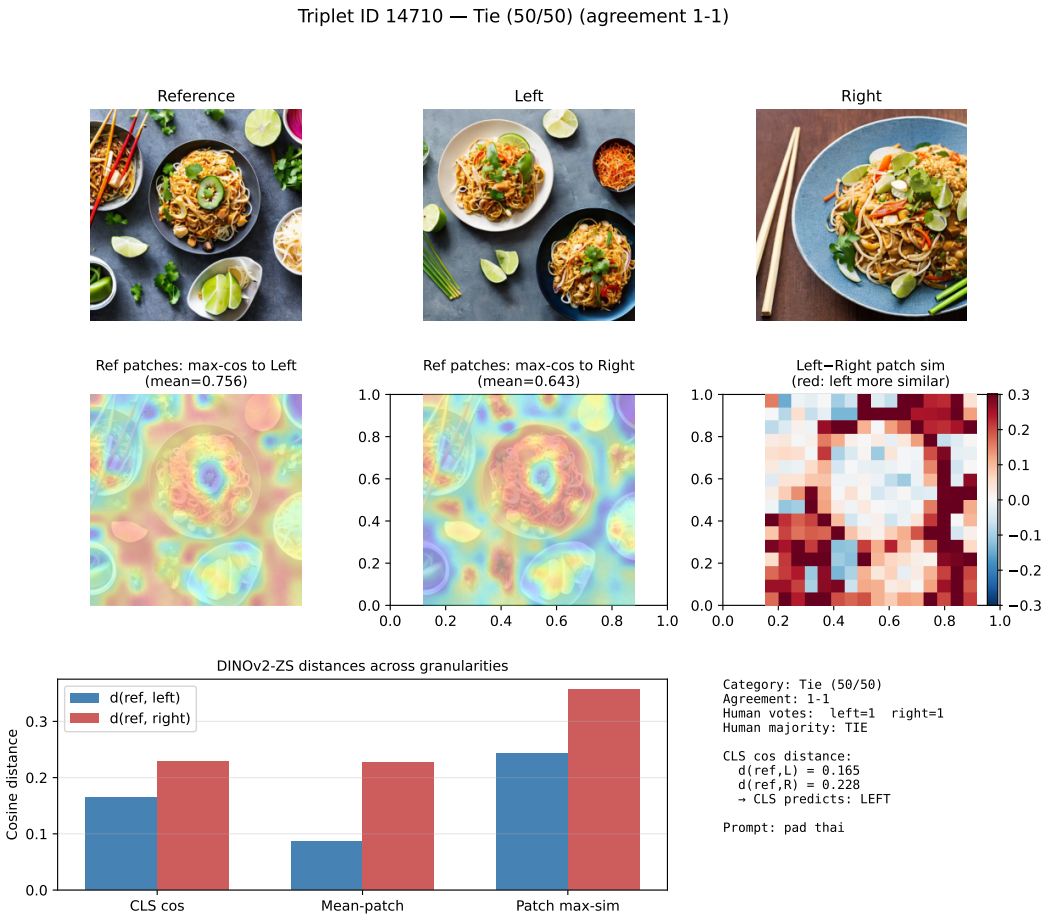


Figure 7: Tie triplet (1-1). Voters split evenly; the released NIGHTS-20k filter discards this case. The patch-similarity heatmaps are nearly identical for the two candidates, the difference map is uniformly low-amplitude, and the two CLS-cosine distances are within tenths of a percent of each other. Tie triplets carry no reproducible labeling signal. In our evaluation, all 35,580 ties are held out and only contribute to the bias audit of Section 3.

E Training details

This section documents the full training recipe for reproducibility. All trained methods in Table 1 use the same pipeline except where explicitly noted; DreamSim-Scratch-100k uses the authors' published training code verbatim and is documented separately at the end.

Preprocessing. Images are decoded from the NIGHTS release JPEGs, resized to 224×224 with bicubic interpolation, and center-cropped. Normalization follows the backbone: ImageNet statistics (mean $[0.485, 0.456, 0.406]$, std $[0.229, 0.224, 0.225]$) for DINOv1, DINOv2, and DINOv3; $0.5/0.5$ for C-RADIOv4; the processor-supplied normalization for DreamSim and Qwen3-VL. No data augmentation is used at training time: the triplet task is orientation-sensitive, and the reference/candidate pairing is the supervision signal.

LoRA configuration. LoRA adapters have rank 8, $\alpha = 16$, and dropout 0.05, and are inserted on the `qkv` and `proj` linear layers of every transformer block. Layer-norm parameters of the backbone are also unfrozen. For DINOv2 ViT-B/14 with 12 blocks this yields 442,368 LoRA parameters plus 1,536 LayerNorm parameters, which is approximately 0.5% of the 86M backbone. The DreamSim ensemble retraining uses rank 16 on the same modules to match the authors' released configuration.

Optimizer. Schedule-Free AdamW [Defazio et al., 2024] with $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-2}, 100 linear warm-up steps, and a constant learning rate thereafter (no cosine schedule). Separate learning rates are used: 10^{-4} for LoRA and layer-norm parameters, 2×10^{-4} for the head. Gradients are clipped to a global norm of 1.0 before each optimizer step.

Batch and compute. Batch size is 32 triplets per step (each triplet contributes three images to the forward pass). Training runs for 10 epochs over the fold's training split. Compute is a shared cluster of NVIDIA RTX A6000 GPUs (48 GB). A full 5-fold run of DINOv2-LoRA-Cog-v3 takes approximately 3–5 hours when five GPUs run concurrently, one per fold.

Checkpoint selection. A held-out validation split of 10% of each fold's training data selects the best-of-10-epochs checkpoint by validation accuracy; the selected checkpoint is evaluated once on the fold's test split, and that single number appears in Table 9. Per-triplet test-set predictions are persisted for every (method, fold) pair so that the Cohen's κ matrix, McNemar tests, and error-distribution analyses can be reproduced without re-running training.

DreamSim retraining from scratch. We run the authors' published training code unchanged with the released ensemble configuration: DINO+CLIP+OpenCLIP, rank-16 LoRA, margin-0.05 hinge loss, batch size 16, 6 epochs. The release filters training triplets to those with at least 6 votes, so the scratch model sees only strong-majority triplets during training; we evaluate on the same full non-tie test fold as every other method in Table 1. This mismatch is the point of the comparison, not an oversight: it isolates what the authors' training recipe would produce when the 100k distribution is available.